

第 35 届全国部分地区大学生物理竞赛

参考解答

2018.12

一、填空题（共 10 题，每题 2 空，每空 3 分，共 60 分）

1. $p + \frac{1}{2}\rho v^2 + \rho gh = \text{常量}$, $\sqrt{2gh_1}$ 。

2. $\sqrt{GM/A}$, $2\pi A\sqrt{A/GM}$ 。

3. $S = k\ln W$, $S_B \geq S_A$ 。

4. 0 , $qQ/4\pi\epsilon_0 R$ 。

5. $\mu_0 Id\vec{l} \times \vec{r}/4\pi r^3$, $\sqrt{2}\mu_0 I/2R$ 。

6. $I_1 + I_2 = I_3$, $I_1 R_1 - I_2 R_2 = \varepsilon_1 - \varepsilon_2$ 。

7. 5.893×10^4 , $K - 200$ 。

8. $I_0/2$, 90° 。

9. 4.5×10^{-3} , 2.0×10^{19} 。

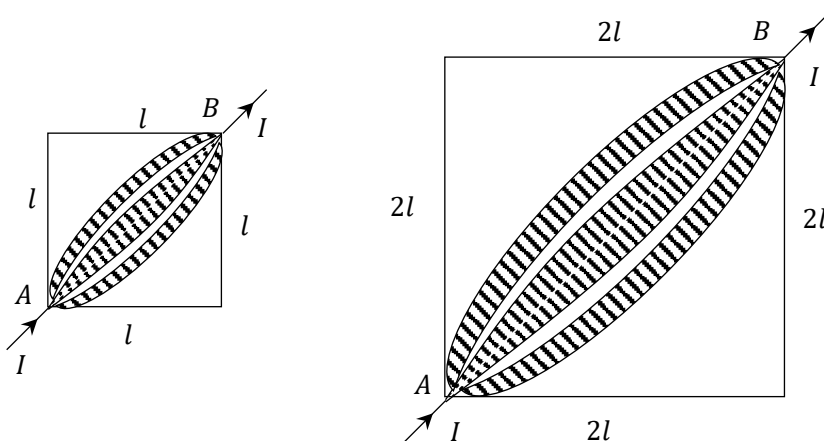
10. l_0/c , $\sqrt{\frac{c-v}{c+v}} \frac{l_0}{c}$ 。

二、计算题（共 4 题，每题 15 分，共 60 分）

11. (15 分)

解：图 1： $R_{AB} = \rho l_3/l_1 l_2$, $R_{AB}^* = \frac{\rho \cdot 2l_3}{2l_1 \cdot 2l_2} = \frac{1}{2}R_{AB} \Rightarrow R_{AB}^* = 5\Omega$ (5 分)

图 2：



右图每一个小流管电阻，会因流管长度加倍而加倍，又会因横向线度也加倍而会降半，故 $R_{AB}^* = R_{AB} = 5\Omega$ 。(10 分)

12. (15 分)

解：(1) 由 $T_A = T_0$ 、 $T_B = 2T_0$ 、 $T_C = 3T_0$ ，可将 A、B、C、D 四处 p 、 V 参量标记为题

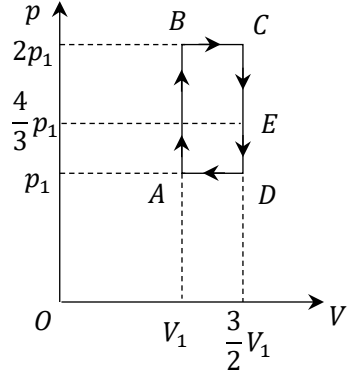
解图 1 所示, 可得 D 处温度和 $C-D$ 过程中存在状态 E , 其状态量分别为

$$T_D = \frac{3}{2}T_0, \quad \begin{cases} p_E = \frac{4}{3}p_1 \\ T_E = 2T_0 \end{cases}$$

据 $U = \nu C_{mV} T$, $C_{mV} = \begin{cases} \frac{5}{2}R, T < 2T_0 \\ \frac{7}{2}R, T \geq 2T_0 \end{cases}$

得 $U-T$ 曲线如题解图 2 所示。 (6 分)

(2)



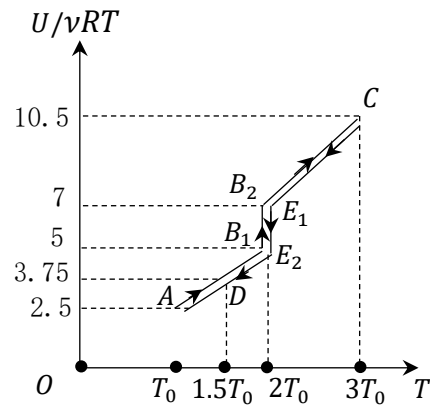
题解图 1

$$Q_{AB\text{吸}} = U_B - U_A = \frac{7}{2}\nu RT_B - \frac{5}{2}\nu RT_A$$

$$= \frac{9}{2}\nu RT_0$$

$$Q_{BC\text{吸}} = \nu C_{mp}^{(1)}(T_C - T_B) = \frac{9}{2}\nu RT_0$$

$$Q_{\text{吸}} = Q_{AB\text{吸}} + Q_{BC\text{吸}} = 9\nu RT_0$$



题解图 2

$$Q_{CD\text{放}} = U_C - U_D = \frac{7}{2}\nu RT_C - \frac{5}{2}\nu RT_D = \frac{27}{4}\nu RT_0$$

$$Q_{DA\text{放}} = \nu C_{mp}^{(2)}(T_D - T_A) = \frac{7}{4}\nu RT_0$$

$$Q_{\text{放}} = Q_{CD\text{放}} + Q_{DA\text{放}} = \frac{34}{4}\nu RT_0$$

得 $\eta = 1 - \frac{Q_{\text{放}}}{Q_{\text{吸}}} = \frac{1}{18} = 5.6\%$ (9 分)

附注: $C_{mp}^{(1)} = \frac{9}{2}R$, $C_{mp}^{(2)} = \frac{7}{2}R$

13. (15 分)

解: 将细杆长记为 $2l$, 每个小球质量记为 m , C 点速度分解已在图中标出。速度间关系如下:

$$v_A \cos \phi = v_B \sin \phi, v_{C\perp} = \frac{1}{2}v_A, v_{C\parallel} = \frac{1}{2}v_B$$

$$v_C^2 = \frac{1}{4}(v_A^2 + v_B^2)$$

由能量守恒, 得

$$\begin{aligned} mg \cdot 2l(1 - \cos \phi) + mgl(1 - \cos \phi) \\ = \frac{1}{2}m(v_A^2 + v_B^2 + v_C^2) \end{aligned}$$

与上式联立可得

$$v_A^2 + v_B^2 = \frac{24}{5}gl(1 - \cos \phi)$$

再将 $v_A = \frac{\sin \phi}{\cos \phi}v_B$ 代入, 得

$$v_B^2 = \frac{24}{5}gl(1 - \cos \phi) \cos^2 \phi$$

系统水平朝右动量

$$P_{\parallel} = mv_B + mv_{C\parallel} = \frac{3}{2}mv_B = \frac{3}{2}m \sqrt{\frac{24}{5}gl(1 - \cos \phi) \cos^2 \phi}$$

ϕ 从 0 增大到 $\frac{\pi}{2}$ 过程中, 函数 $(1 - \cos \phi) \cos^2 \phi$ 乃至 $P_{\parallel}$ 都从 0 增大, 但中间存在极大值。

$P_{\parallel}$ 的增大需由墙面给 A 端的水平朝右支持力 N 提供的冲量来实现。当 $P_{\parallel}$ 增到极大值时 $N = 0$, A 端需离开墙面。可得结论:

A 端未达墙的底端 O 之前, A 端已离开墙面。 (9 分)

由

$$(1 - \cos \phi) \cos^2 \phi = 4(1 - \cos \phi) \left(\frac{1}{2} \cos \phi\right) \left(\frac{1}{2} \cos \phi\right)$$

和代数不等式

$$ABC \leq \frac{1}{3^3}(A + B + C)^3, A、B、C \text{ 均为正, 等号在 } A = B = C \text{ 时取得}$$

可知, 当

$$1 - \cos \phi = \frac{1}{2} \cos \phi = \frac{1}{2} \cos \phi, \text{ 即 } \cos \phi = \frac{2}{3}$$

时, $(1 - \cos \phi) \cos^2 \phi$ 取得极大值, 即 $P_{\parallel}$ 取得极大值。据此,

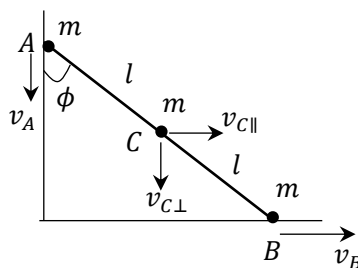
$$\phi = \arccos \frac{2}{3} = 48.2^\circ$$

时, A 端离开墙面。(6 分)

14. (15 分)

解: (1) 求 α :

$$n(y_0) = \alpha n_0 \sqrt{\frac{A_0^2}{16 \cdot \frac{3}{16} A_0^2} + 5} = \alpha n_0 \sqrt{\frac{1}{3} + 5} = \alpha n_0 \frac{4}{\sqrt{3}}$$



将 $n(y_0) = n_0$ 代入, 即得

$$\alpha = \frac{\sqrt{3}}{4} \quad (3 \text{ 分})$$

(2) 求光线方程:

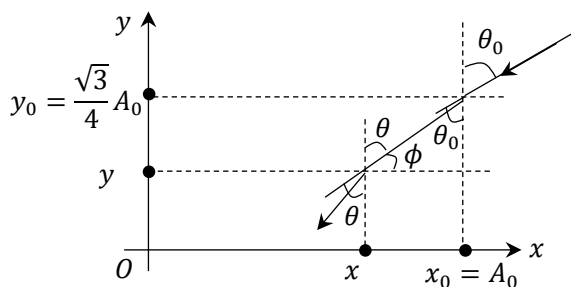
参考题解图, 有

$$\cos \phi = \sin \theta = \frac{n_0}{n(y)} \sin \theta_0$$

$$= \frac{n_0}{\alpha n_0 \sqrt{\frac{A_0^2}{16y^2} + 5}} \cdot \frac{\sqrt{3}}{2}$$

$$= \frac{\frac{\sqrt{3}}{2}}{\frac{\sqrt{3}}{4} \sqrt{\frac{A_0^2}{16y^2} + 5}}$$

$$= \frac{2}{\frac{1}{4} \sqrt{\frac{A_0^2}{y^2} + 80}} = \frac{8}{\sqrt{\frac{A_0^2}{y^2} + 80}}$$



题解图

$$\Rightarrow \cos^2 \phi = \frac{64}{\frac{A_0^2}{y^2} + 80}$$

又有

$$\left(\frac{dy}{dx}\right)^2 = (\tan \phi)^2 = \frac{\sin^2 \phi}{\cos^2 \phi} = \frac{1 - \cos^2 \phi}{\cos^2 \phi} = \frac{1}{\cos^2 \phi} - 1$$

$$= \frac{\frac{A_0^2}{y^2} + 80}{64} - 1 = \left(\frac{A_0^2}{64y^2} + \frac{64}{64} + \frac{16}{64}\right) - 1 = \frac{A_0^2}{64y^2} + \frac{1}{4}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{1}{4} \left(\frac{A_0^2}{16y^2} + 1\right) = \frac{1}{4} \frac{A_0^2 + 16y^2}{16y^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{A_0^2 + 16y^2}}{8y}$$

$$\Rightarrow dx = \frac{8y}{\sqrt{A_0^2 + 16y^2}} dy$$

引入参量 u , 使得

$$u^2 = A_0^2 + 16y^2 \Rightarrow 2udu = 32ydy \Rightarrow 8ydy = \frac{1}{2}udu$$

代入上式, 得

$$dx = \frac{\frac{1}{2}udu}{u} = \frac{1}{2}du$$

$$x_0 = A_0 \text{ 对应 } y_0 = \frac{\sqrt{3}}{4}A_0, \Rightarrow u_0^2 = A_0^2 + 16y_0^2 = A_0^2 + 3A_0^2 = 4A_0^2 \Rightarrow u_0 = 2A_0$$

与积分式

$$\int_{x_0}^x dx = \int_{u_0}^u \frac{1}{2} du$$

联立, 得

$$\begin{aligned} x - x_0 &= \frac{1}{2}(u - u_0) \Rightarrow x - A_0 = \frac{1}{2}(u - 2A_0) = \frac{1}{2}u - A_0 \\ \Rightarrow x &= \frac{1}{2}u = \frac{1}{2}\sqrt{A_0^2 + 16y^2} \Rightarrow 4x^2 = A_0^2 + 16y^2 \\ \Rightarrow 4x^2 - 16y^2 &= A_0^2 \end{aligned}$$

所求光线方程为

$$\frac{x^2}{\left(\frac{A_0}{2}\right)^2} - \frac{y^2}{\left(\frac{A_0}{4}\right)^2} = 1$$

即为双曲线。(12 分)

15. (20 分)

解: (1)

据(A)式可得 (1.1) 问解答:

$$u^2 = \left[1 - \frac{1}{(1+\alpha x)^2}\right] c^2 \Rightarrow u = \frac{\sqrt{\alpha x(2+\alpha x)}}{1+\alpha x} c \quad (1) \quad (2 \text{ 分})$$

据(B)式可得 (1.2) 问解答:

$$u^2 = \frac{\alpha^2 c^4 t^2}{(1+\alpha^2 c^2 t^2)} \Rightarrow u = \frac{\alpha c^2 t}{\sqrt{1+\alpha^2 c^2 t^2}} \quad (2) \quad (2 \text{ 分})$$

联立(1)、(2)式或联立左侧两式, 可得 (1.3) 问解答: (2 分)

$$x = \frac{1}{\alpha}(\sqrt{1+\alpha^2 c^2 t^2} - 1) \quad (3)$$

也可写出逆向关系式: $t = \sqrt{\frac{(2+\alpha x)x}{\alpha c^2}} \quad (3)'$

(2)

质点在切向恒力作用下沿圆周切线方向运动的过程中, 功能关系、冲量—动量关系, 类似于在 x 轴方向恒力作用下沿 x 轴方向运动的过程中, 功能关系、冲量—动量关系。故有

$$\bar{N} = \frac{\int_0^T N dt}{T} = \frac{\int_0^T m \frac{u^2}{R} dt}{T} = \frac{\int_0^T m u \frac{u dt}{R}}{T}, \quad T: \text{运动总时间}$$

方案 I: $Fx = mc^2 - m_0 c^2 \Rightarrow m = \frac{Fx}{c^2} + m_0, udt = dx$

$$\Rightarrow \bar{N} = \frac{1}{RT} \int_0^{2\pi R} \left(\frac{Fx}{c^2} + m_0\right) u dx \quad u = \frac{\sqrt{\alpha x(2+\alpha x)}}{1+\alpha x} c$$

方案 II:

$$\begin{aligned} Ft &= mu, \quad u = \frac{\alpha c^2 t}{\sqrt{1+\alpha^2 c^2 t^2}} \\ \Rightarrow \bar{N} &= \frac{\int_0^T m u \frac{u dt}{R}}{T} = \frac{F}{RT} \int_0^T \frac{t \alpha c^2 t}{\sqrt{1+\alpha^2 c^2 t^2}} dt \\ &= \frac{F}{\alpha RT} \int_0^T \frac{\alpha^2 c^2 t^2 dt}{\sqrt{1+\alpha^2 c^2 t^2}} \quad (2 \text{ 分}) \end{aligned}$$

引入辅助量

$$\begin{aligned}\beta = 1 + \alpha^2 c^2 t^2 &\Rightarrow t^2 = \frac{\beta - 1}{\alpha^2 c^2}, \quad t = \frac{\sqrt{\beta - 1}}{\alpha c}, \quad d\beta = 2c^2 \alpha^2 t dt \\ &\Rightarrow d\beta = 2\alpha^2 c^2 \frac{\sqrt{\beta - 1}}{\alpha c} dt = 2\alpha c \sqrt{\beta - 1} dt \\ &\Rightarrow dt = \frac{d\beta}{2\alpha c \sqrt{\beta - 1}}\end{aligned}$$

得

$$\begin{aligned}\bar{N} &= \frac{F}{\alpha RT} \int_0^T \frac{\beta - 1}{\sqrt{\beta}} \cdot \frac{d\beta}{2\alpha c \sqrt{\beta - 1}} = \frac{F}{2\alpha^2 c RT} \int_0^T \frac{\sqrt{\beta - 1} d\beta}{\sqrt{\beta}} \\ &\Rightarrow \bar{N} = \frac{F}{2\alpha^2 c RT} \int_0^T \sqrt{1 - \frac{1}{\beta}} d\beta \quad (C)\end{aligned}$$

引入

$$\begin{aligned}\gamma = \sqrt{\beta} &\Rightarrow d\gamma = \frac{1}{2} \frac{d\beta}{\sqrt{\beta}} \Rightarrow d\beta = 2\sqrt{\beta} d\gamma = 2\gamma d\gamma \\ &\Rightarrow \int_0^T \sqrt{1 - \frac{1}{\beta}} d\beta = \int_0^T \sqrt{1 - \frac{1}{\gamma^2}} 2\gamma d\gamma = 2 \int_0^T \sqrt{\gamma^2 - 1} d\gamma\end{aligned}$$

因

$$\begin{aligned}\int \sqrt{\gamma^2 - 1} d\gamma &= \frac{\gamma}{2} \sqrt{\gamma^2 - 1} - \frac{1}{2} \ln(\gamma + \sqrt{\gamma^2 - 1}) + C \\ &\Rightarrow \int_0^T \sqrt{1 - \frac{1}{\beta}} d\beta = \left[\gamma \sqrt{\gamma^2 - 1} - \ln(\gamma + \sqrt{\gamma^2 - 1}) \right]_0^T, \quad \gamma = \sqrt{\beta} \\ &= \left[\sqrt{\beta} \sqrt{\beta - 1} - \ln(\sqrt{\beta} + \sqrt{\beta - 1}) \right]_0^T, \quad \beta = 1 + \alpha^2 c^2 t^2 \\ &\Rightarrow \int_0^T \sqrt{1 - \frac{1}{\beta}} d\beta = \left[\sqrt{1 + \alpha^2 c^2 t^2} \sqrt{\alpha^2 c^2 t^2} - \ln(\sqrt{1 + \alpha^2 c^2 t^2} + \sqrt{\alpha^2 c^2 t^2}) \right]_0^T \\ &= \left[\sqrt{1 + \alpha^2 c^2 T^2} \cdot \alpha c T - \ln(\sqrt{1 + \alpha^2 c^2 T^2} + \alpha c T) \right]\end{aligned}$$

代入(C)式, 得

$$\bar{N} = \frac{F}{2\alpha^2 c RT} \left[\sqrt{1 + \alpha^2 c^2 T^2} \cdot \alpha c T - \ln(\sqrt{1 + \alpha^2 c^2 T^2} + \alpha c T) \right]$$

其中, 由(3)'式得

$$T = \sqrt{\frac{(2 + \alpha x)x}{\alpha c^2}} \Big|_{x=2\pi R} = \sqrt{\frac{(2 + 2\pi R\alpha)2\pi R}{\alpha c^2}}$$

代入

$$\alpha = \frac{1}{\pi R}$$

得

$$\begin{aligned}\bar{N} &= \frac{F}{2\alpha^2 cRT} \left[\sqrt{1 + \alpha^2 c^2 T^2} \cdot \alpha cT - \ln \left(\sqrt{1 + \alpha^2 c^2 T^2} + \alpha cT \right) \right] \\ 2\alpha^2 cRT &= \frac{4\sqrt{2}}{\pi}, \quad \sqrt{1 + \alpha^2 c^2 T^2} = 3, \quad \alpha cT = 2\sqrt{2} \\ \Rightarrow \bar{N} &= \frac{\pi F}{4\sqrt{2}} [3 \cdot 2\sqrt{2} - \ln(3 + 2\sqrt{2})] \\ \Rightarrow \sigma \frac{\pi}{4\sqrt{2}} [3 \cdot 2\sqrt{2} - \ln(3 + 2\sqrt{2})] &= 3.73 \quad (12 \text{ 分})\end{aligned}$$

16.

解：基本方程组（参考题解图）

地面系：M 运动： $F + f \cos \phi - N \sin \phi = Ma_M$ (1)

M 系：m 运动： $mg \sin \phi - f - ma_M \cos \phi = ma_m$ (2)

$N = mg \cos \phi + ma_M \sin \phi$ (3) (3 分)

$\left. \begin{matrix} fR = I\beta = \frac{2}{5}mR^2\beta \\ \beta R = a_m \end{matrix} \right\} f = \frac{2}{5}ma_m$ (4) (2 分)

(1) $a_m = 0$ 时，必有 $f = 0$ ，由(2)式即得

$$a_M(1) = g \tan \phi \quad (2 \text{ 分})$$

(2) m 相对 M 不动时，m 与 M 一起朝右运动，故(1)

问对应 F_0 的为

$$F_0 = (M + m)a_M(1) = (M + m)g \tan \phi \quad (1 \text{ 分})$$

(附注：由基本方程组(3)得

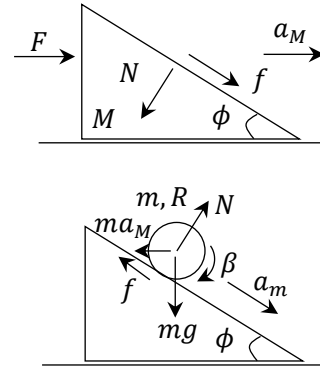
$$\begin{aligned}N &= mg \cos \phi + ma_M(1) \sin \phi = mg \cos \phi + mg \frac{\sin \phi}{\cos \phi} \sin \phi \\ \Rightarrow N \sin \phi &= mg \cos \phi \sin \phi + mg \frac{\sin \phi}{\cos \phi} \cdot \sin^2 \phi \\ &= mg \frac{\sin \phi}{\cos \phi} (\cos^2 \phi + \sin^2 \phi) = mg \frac{\sin \phi}{\cos \phi} = ma_M(1)\end{aligned}$$

代入基本方程组(1)，也可得

$$F = N \sin \phi + Ma_M(1) = ma_M(1) + Ma_M(1) = (M + m)a_M(1) \quad)$$

将基本方程组(3)、(4)式代入(1)、(2)式，得

$$\begin{aligned}&\begin{cases} F + \frac{2}{5}m \cos \phi a_m - mg \sin \phi \cos \phi - m \sin^2 \phi a_M = Ma_M \\ mg \sin \phi - \frac{2}{5}ma_m - ma_M \cos \phi = ma_m \end{cases} \\ \Rightarrow &\begin{cases} \frac{2}{5}m \cos \phi a_m = -F + mg \sin \phi \cos \phi + m \sin^2 \phi a_M + Ma_M \\ \frac{7}{5}ma_m = mg \sin \phi - m \cos \phi \cdot a_M \end{cases} \quad (5) \\ \Rightarrow &\frac{2 \cos \phi}{7} = \frac{-F + mg \sin \phi \cos \phi + (M + m \sin^2 \phi)a_M}{mg \sin \phi - m \cos \phi \cdot a_M} \\ \Rightarrow &2mg \sin \phi \cos \phi - 2m \cos^2 \phi \cdot a_M = -7F + 7mg \sin \phi \cos \phi + 7(M + m \sin^2 \phi)a_M \\ \Rightarrow &(7M + 5m \sin^2 \phi + 2m)a_M = 7F - 5mg \sin \phi \cos \phi\end{aligned}$$



题解图

$$\Rightarrow a_M = \frac{7F - 5mg \sin \phi \cos \phi}{7M + 5m \sin^2 \phi + 2m} \quad (6 \text{ 分})$$

将 $F = 2F_0 = 2(M + m)g \tan \phi$ 代入, 即得

$$a_M(2) = \frac{14(M + m)g \tan \phi - 5mg \sin \phi \cos \phi}{7M + 5m \sin^2 \phi + 2m}$$

由(5)式, 得

$$\begin{aligned} \frac{7}{5}ma_m &= mg \sin \phi - m \cos \phi \frac{14(M + m)g \tan \phi - 5mg \sin \phi \cos \phi}{7M + 5m \sin^2 \phi + 2m} \\ &= \frac{7Mmg \sin \phi + 5m^2g \sin^3 \phi + 2m^2g \sin \phi - 14(M + m)mg \sin \phi + 5m^2g \sin \phi \cos^2 \phi}{7M + 5m \sin^2 \phi + 2m} \\ &= \frac{\frac{7}{5}m(7M + 5m \sin^2 \phi + 2m)}{mg \sin \phi} a_m = 7M + 5m \sin^2 \phi + 2m + 5m \cos^2 \phi - 14(M + m) \\ &= -7(M + m) < 0 \\ &\Rightarrow a_m < 0 \end{aligned}$$

故小球沿斜面向上滚动。

(6 分)

(附注:

$$a_m = -\frac{5(M+m) \sin \phi}{7M + 5m \sin^2 \phi + 2m} g \quad)$$

17. (20 分)

解: t 时刻感应电动势记为 ε , 有

$$Blv = \varepsilon = \frac{q}{C} + iR = \frac{q}{C} + R \frac{dq}{dt} \quad (1)$$

$$\Rightarrow \frac{dv}{dt} = \frac{R}{Bl} \frac{d^2q}{dt^2} + \frac{dq/dt}{BlC} \quad (2)$$

棒的运动方程为

$$m \frac{dv}{dt} = mg - iBl = mg - Bl \frac{dq}{dt}$$

将(2)式代入, 得

$$\begin{aligned} \frac{mR}{Bl} \frac{d^2q}{dt^2} &= mg - Bl \frac{dq}{dt} - \frac{m}{BlC} \frac{dq}{dt} = mg - \left(Bl + \frac{m}{BlC} \right) \frac{dq}{dt} \\ \Rightarrow \frac{d^2q}{dt^2} &= \frac{Blg}{R} - \left(\frac{B^2l^2}{mR} + \frac{1}{RC} \right) \frac{dq}{dt} = \frac{Blg}{R} - \frac{2}{RC} \frac{dq}{dt} \\ &\Rightarrow \frac{d^2q}{dt^2} + \frac{2}{RC} \frac{dq}{dt} = \frac{Blg}{R} \\ \Rightarrow \frac{dq}{dt} &= e^{\int -\frac{2}{RC} dt} \left(\int \frac{Blg}{R} \cdot e^{\frac{2}{RC} dt} dt + A \right) \\ &= e^{-\frac{2}{RC} t} \left(\int \frac{Blg}{R} \cdot e^{\frac{2}{RC} t} dt + A \right) \\ &= e^{-\frac{2}{RC} t} \left(\frac{RC}{2} \frac{Blg}{R} e^{\frac{2}{RC} t} + A \right) \\ &= \frac{\sqrt{mC}g}{2} + Ae^{-\frac{2}{RC} t} \end{aligned}$$

由 $t = 0$ 时, $dq/dt = i = 0$, 得

$$\begin{aligned}
 A &= -\frac{\sqrt{mC}g}{2} \Rightarrow \frac{dq}{dt} = \frac{\sqrt{mC}g}{2} \left(1 - e^{-\frac{2}{RC}t}\right) \\
 &\Rightarrow \int_0^q dq = \frac{\sqrt{mC}g}{2} \int_0^t \left(1 - e^{-\frac{2}{RC}t}\right) dt \\
 &= \frac{\sqrt{mC}g}{2} \left(t + \frac{RC}{2} e^{-\frac{2}{RC}t} \Big|_0^t\right) = \frac{\sqrt{mC}g}{2} \left(t + \frac{RC}{2} e^{-\frac{2}{RC}t} - \frac{RC}{2}\right)
 \end{aligned}$$

得 $q \sim t$ 关系:

$$q = \frac{\sqrt{mC}g}{2} \left[t - \frac{RC}{2} \left(1 - e^{-\frac{2}{RC}t}\right)\right] \quad (15 \text{ 分})$$

代入(1)式, 得 $v \sim t$ 关系:

$$\begin{aligned}
 v &= \frac{q}{BlC} + \frac{R}{Bl} \frac{dq}{dt} = \frac{g}{2} \left[t - \frac{RC}{2} \left(1 - e^{-\frac{2}{RC}t}\right)\right] + \frac{RCg}{2} \left(1 - e^{-\frac{2}{RC}t}\right) \\
 &= \frac{gt}{2} + \frac{RCg}{4} \left(1 - e^{-\frac{2}{RC}t}\right)
 \end{aligned}$$

有

$$t = 0 \text{ 时}, \quad v = 0; \quad t \rightarrow \infty, \quad v \rightarrow \infty \quad (5 \text{ 分})$$